

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(b)	(i)	Reactance of inductor equal and opposite to capacitance (or pds equal and opposite) (1) Minimum impedance (or all pd across resistor) (1) Maximum current (1)	3			3		
		(ii)	Algebra leading to $f = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$ (1) Answer = 130 kHz (1)		2		2	2	
		(iii)	$X_L = 65\,000\, [\Omega]$ or $X_C = 10\,500\, [\Omega]$ (1) Substitution into $Z = \sqrt{(X_L - X_C)^2 + R^2} = 55\,000\, \Omega$ (1) Use of $I = \frac{V}{Z}$ leading to Answer = 91 μA (ecf) (1)		3		3	3	
		(iv)	Smallest capacitor used (or trial and error) (1) Q factor calculated (533) (or other valid method of getting V) (1) Pd across C or L = $Q \times 5$ (2 700 V) (1) Peak pd this is $\times \sqrt{2}$ (3 800 V) (1) Student is wrong, it's considerably larger (valid conclusion) (1)			5	5	3	
			Question 6 total	6	9	5	20	10	0